

Tharun R

M.Tech (Integrated) Computer Science and Engineering
Vellore Institute of Technology, Vellore — Expected Graduation: 2027
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Objective

M.Tech CSE student skilled in Python, NLP, and LLM-based applications. Experienced in building RAG-based systems and REST APIs. Hands-on with FastAPI and OpenAI APIs, with strong foundations in DSA and system design.

Education

- **Vellore Institute of Technology, Vellore** *Expected: 2027*
Full-time M.Tech (Integrated) Computer Science and Engineering — **CGPA: 9.11**
- **Rishi Matriculation Higher Secondary School, Ranipet** *2022*
HSC (12th Standard) — **Percentage: 90.33%** (School 1st Rank)
- **Rishi Matriculation Higher Secondary School, Ranipet** *2020*
SSLC (10th Standard) — **Percentage: 89%**

Technical Skills

- **Programming:** Python (Advanced), Java, C++, JavaScript
- **Core CS:** Data Structures & Algorithms, OOP, System Design
- **Backend & APIs:** FastAPI, Flask, REST APIs, Async Programming (asyncio), Postman API
- **AI Engineering & GenAI:** LLMs (OpenAI API), RAG Pipelines, Prompt Engineering, Embeddings, Vector Search
- **Caching & Optimization:** Redis (Basics), In-memory caching, Performance Optimization
- **Databases:** PostgreSQL, MySQL, MongoDB, SQL Optimization
- **Machine Learning & Frameworks:** Deep Learning, Reinforcement Learning, Scikit-learn, PyTorch, NLP, TensorFlow
- **Tools & Platforms:** Docker (Basics), Git, Linux, Power BI, Tableau, Google Analytics
- **Concepts:** Multithreading, Multiprocessing, ETL Pipelines
- **Soft Skills:** Problem solving, Critical Thinking, Team collaboration, Adaptability, Communication

Projects

- **Hospital Management System with Agentic AI** *Python, REST APIs, LLM*
 - Built a hospital management platform to automate scheduling, patient record handling, and resource allocation.
 - Integrated an LLM-based chatbot using Telegram APIs for real-time query handling
 - Designed API-driven backend architecture to manage workflows and enable AI agent-based automation.
- **ResuScope: AI Resume Analyzer** *Flask, NLP, Gemini API*
 - Developed an AI-based resume analysis system using LLMs and semantic similarity models.
 - Implemented embedding-based matching for resume-job description alignment.
 - Built REST APIs to process resumes and generate structured feedback.
- **Cardiac Arrhythmia Detection System** *Python, Artificial Neural Networks*
 - Developed an ANN model to classify ECG signals and detect cardiac arrhythmia patterns.
 - Applied signal preprocessing and feature extraction techniques for improved classification accuracy.
 - Implemented data normalization and model evaluation using performance metrics such as accuracy, precision, and recall.
- **Blockchain Certificate Verification System** *Solidity, IPFS, Web3*
 - Developed a blockchain-based application for secure certificate issuance and verification.
 - Implemented Solidity smart contracts to ensure tamper-proof certificate validation.
 - Integrated QR-based certificate verification with decentralized storage.

Certifications

- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate
- Postman API Fundamentals Student Expert
- AWS Solution Architect Associate and Cloud Practitioner Certification course
- Tata - GenAI Powered Data Analytics Job Simulation
- Deep Learning and Generative AI Fundamentals